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Assessing Pressures Shaping Natural Language Lexica

Jeanne Bruneau--Bongard,^{a,b}  Emmanuel Chemla,^{a,c} 
Thomas Brochhagen^b 

^a*Département d'Études Cognitives, Laboratoire de Sciences Cognitives et Psycholinguistique,
ENS, PSL University*

^b*Department of Translation and Language Sciences, Universitat Pompeu Fabra*

^c*Earth Species Project, Berkeley, CA*

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Abstract

Human languages balance communicative informativity with complexity, conveying as much as needed through the simplest means required to do so. Yet, these concepts—informativity and complexity—have been operationalized in various ways, and it remains unclear which definitions best capture empirical linguistic patterns. A particularly successful operationalization is that offered by the Information Bottleneck framework, which suggests a balance between complexity and informativity across domains like color, kinship, and number. However, we show that the notion of complexity employed by this framework has some counterintuitive consequences. Focusing on color terms, we then study to what extent this and other notions of complexity play a role in explaining cross-linguistic regularity. We propose a method to assess their explanatory contributions; and to probe whether they enter in a joint optimization or in a trade-off competition. This offers a more general framework to study language change and the forces that shape it, where instead of showing that a given model is compatible with existing data, the data is used to adjudicate between candidate measures.

Keywords: Information theory; Language complexity; Semantic typology; Computational linguistics; Language change

1. Current models of language change

Languages across the world carve up reality in different ways. For instance, some languages distinguish between colors that other languages amalgamate into a single word. Such

Correspondence should be sent to Jeanne Bruneau--Bongard, Laboratoire de Sciences Cognitives et Psycholinguistique, Département d'Études Cognitives, ENS, PSL University, EHESS, CNRS, 29 rue d'Ulm, 75005 Paris, France. E-mail: jeanne.bruneau--bongard@ens.psl.eu